

Energy Price Modelling: A Comparative Evaluation of Four Generations of Forecasting Methods

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03.06.2026

Motivation

Importance of accurate Electricity Price Forecasts:

- Optimization of resource allocation
- Minimization of Production Costs
- Risk mitigation for stakeholders across energy supply chain

Research Studies dealing with electricity price forecasting:

- Set of covered models: do not include Transformer-based neural networks (Xu et al., 2023), (Cantillo-Luna et al., 2023), (Zhong et al., 2023), (Zhang et al., 2023)
- Six other papers focused on Machine Learning techniques (no transformers) prior to 2022

Motivation

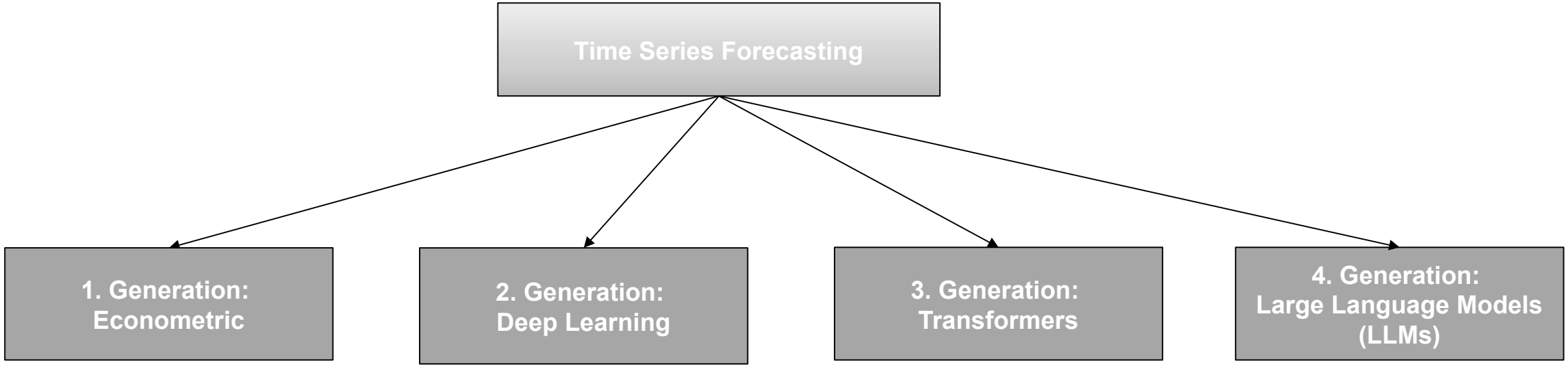
Recently introduced Transformer networks:

- Widely adopted in long-term time series forecasting
- Performance of novel frameworks examined on the same test subset of well-known benchmark time series

Contributions of our Research:

- Comparative empirical analysis of four generations of time series forecasting techniques with the focus on transformers
- Long-term forecasting performance examined on so far unexplored electricity price time series
- Walk-forward validation approach for a robust performance estimation
- Implementing 4th generation Foundational Models (FM): TimesFM and Chronos

Generations of Time Series Forecasting Methods



- Simple linear models

- Memory state models
- Kernel (convolutional) based models

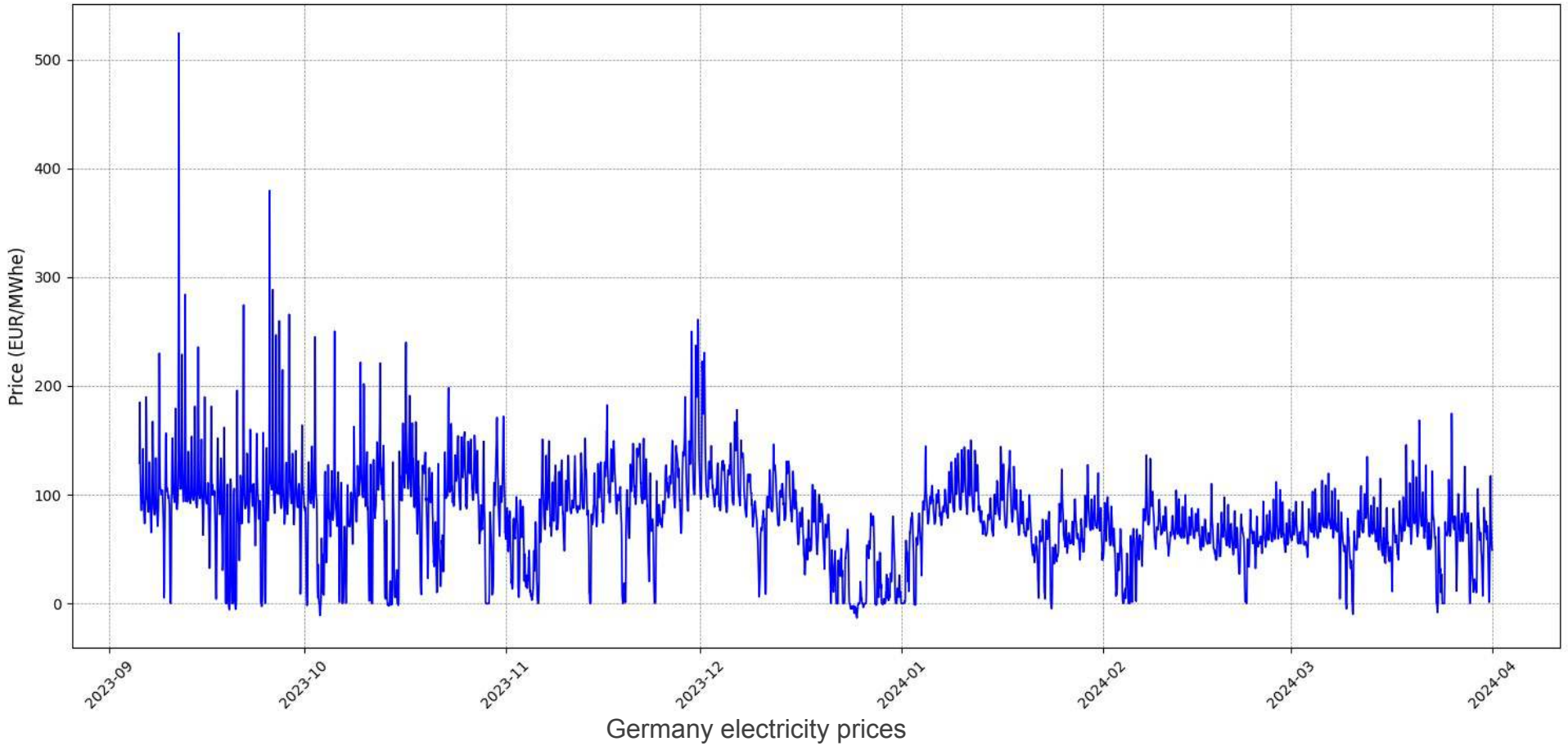
- Encoder-Decoder models
- Specifically designed for long-term forecasting

- Pre-trained language models adapted to time series
- Zero-shot forecasting

Generations of Time Series Forecasting Methods

Time Series Model (Reference)	Gen.	Forecasting Approach						Methodology		Dataset		Walk-Foward Performance Validation
		Econ.	Deep Learning			Transsformer	LLM	Decomposition	Novel Self-Attention Module	Electricity Prices	Synthetic Dataset	
			MLP	CNN	RNN							
Dlinear (Zeng et al., 2022)	2	-	✓	-	-	✓	-	✓	-	-	-	-
Nlinear (Zeng et al., 2022)	2	-	✓	-	-	✓	-	-	-	-	-	-
TSMixer (S.-A. Chen et al., 2023)	2	✓	✓	-	-	✓	-	-	-	-	-	-
Autoformer (Wu et al., 2021)	3	✓	-	-	-	✓	-	✓	✓	-	-	-
Basisformer (Ni et al., 2023)	3	-	✓	✓	-	✓	-	-	✓	-	-	-
Informer (H. Zhou et al., 2021)	3	✓	-	-	✓	-	-	-	✓	-	-	-
PatchTST (Nie, Nguyen, et al., 2024)	3	-	-	✓	✓	✓	-	✓	✓	-	-	-
Quatformer (W. Chen et al., 2022)	3	✓	✓	-	-	✓	✓	-	-	-	✓	-
Chronos (Ansari et al., 2024)	4	-	✓	-	-	✓	✓	-	-	-	✓	-
TimesFM (Das et al., 2024)	4	✓	✓	-	✓	✓	✓	-	-	-	✓	-

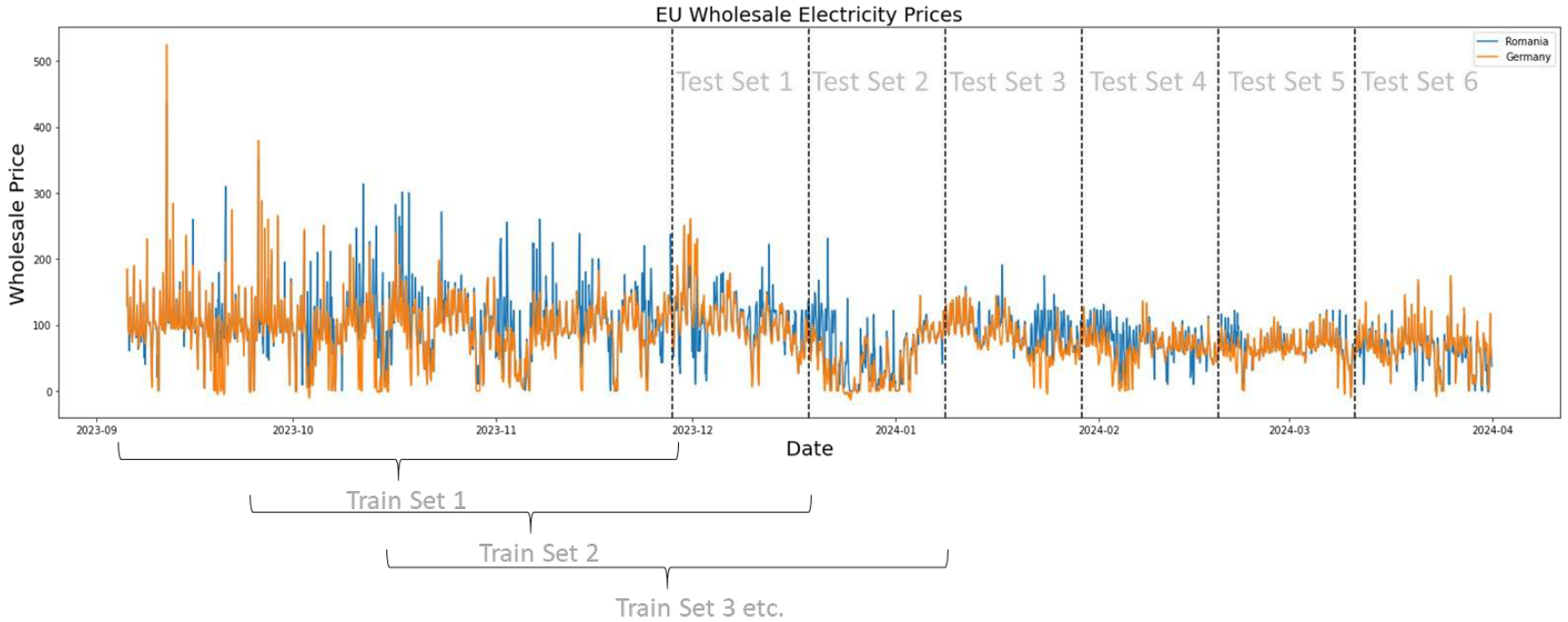
Electricity Price Dataset



Electricity Price Dataset

Wholesale electricity prices for 27 EU countries retrieved from the Ember Climate platform (originally sourced from ENTSO-e).

Walk-forward validation with sliding window approach



Walk-forward validation with a rolling window based on the electricity prices of Romania and Germany

Experimental Setup

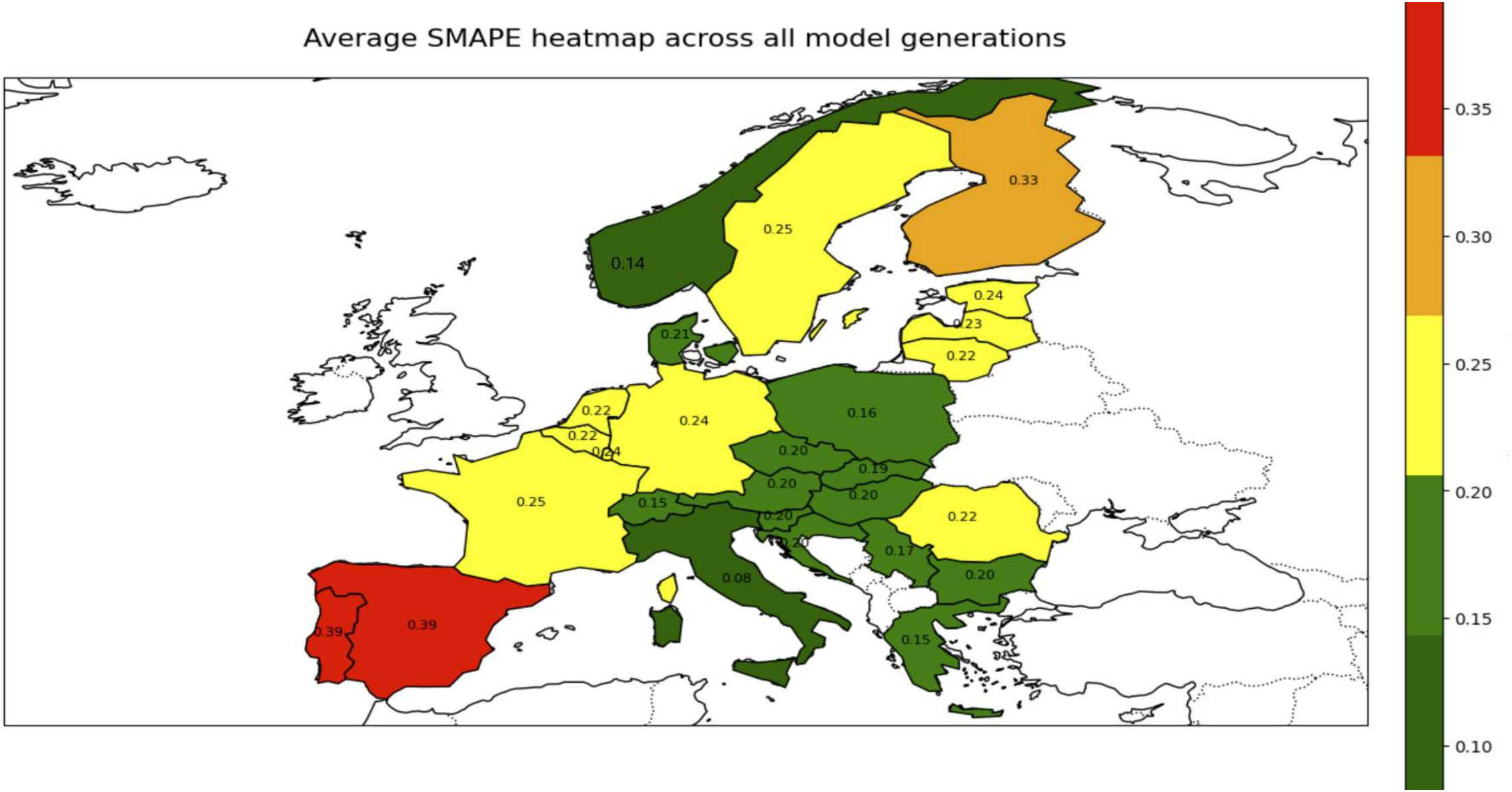
Model training:

- Input sequence length : 96 time steps
- Prediction sequence length: 96 time steps
- Number of training epochs: default values, e.g., 10 epochs
- Early Stopping Callback: 1 epoch of no improvement on train subset
- Transfer learning of trained models to every following train subset
- Walk-forward validation: each train subset consisting of 2000 hours, i.e., 83 days , each test subset of 500 hours, i.e., 20 days

Performance Evaluation:

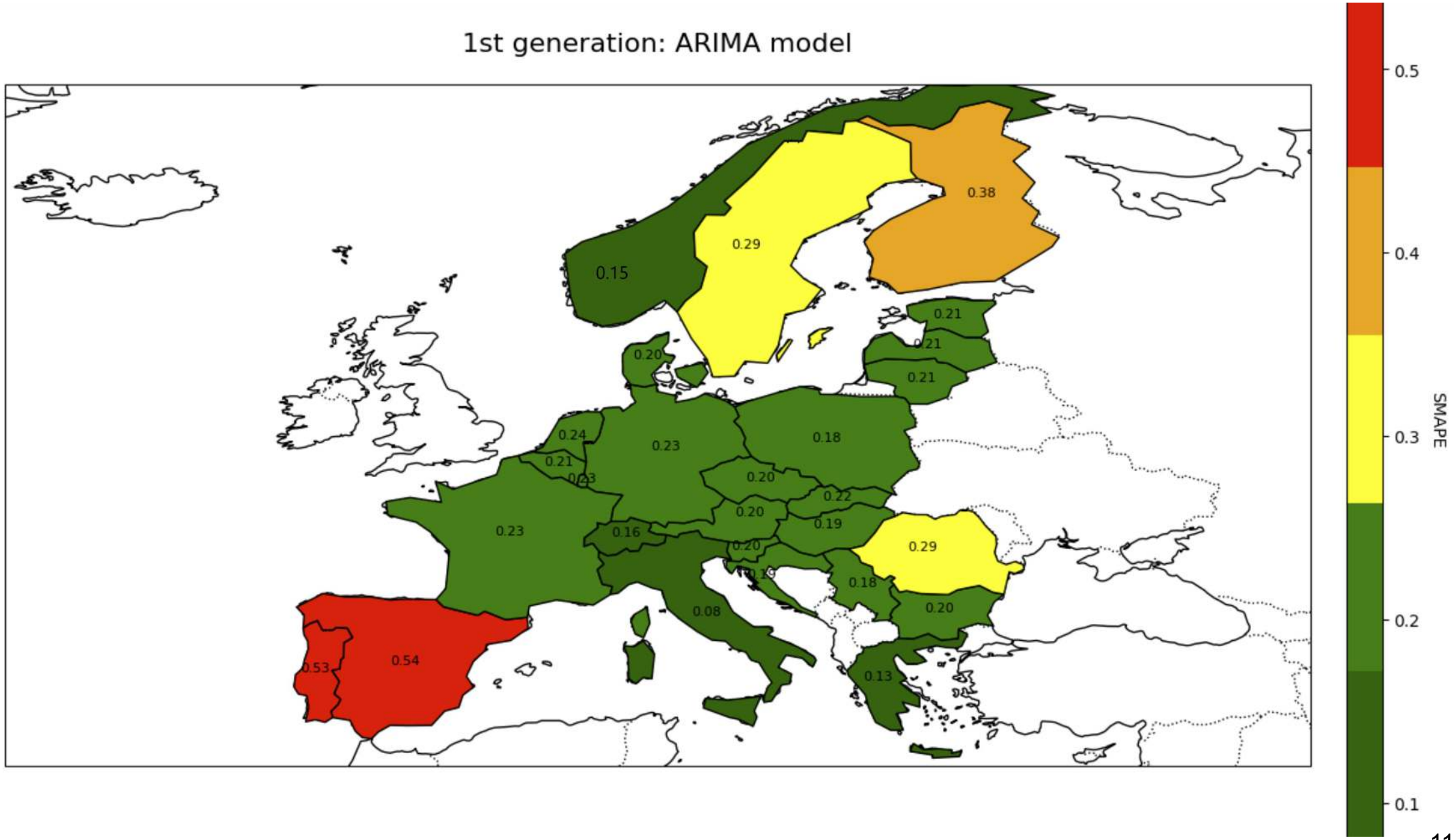
- RMSE, SMAPE
- Different aspects of long-term forecast evaluated: overall distribution of achieved errors, average performance per test subset, and per country
- Composite indicator PI = (100·SMAPE + RMSE) / 2 used for ranking.

Results

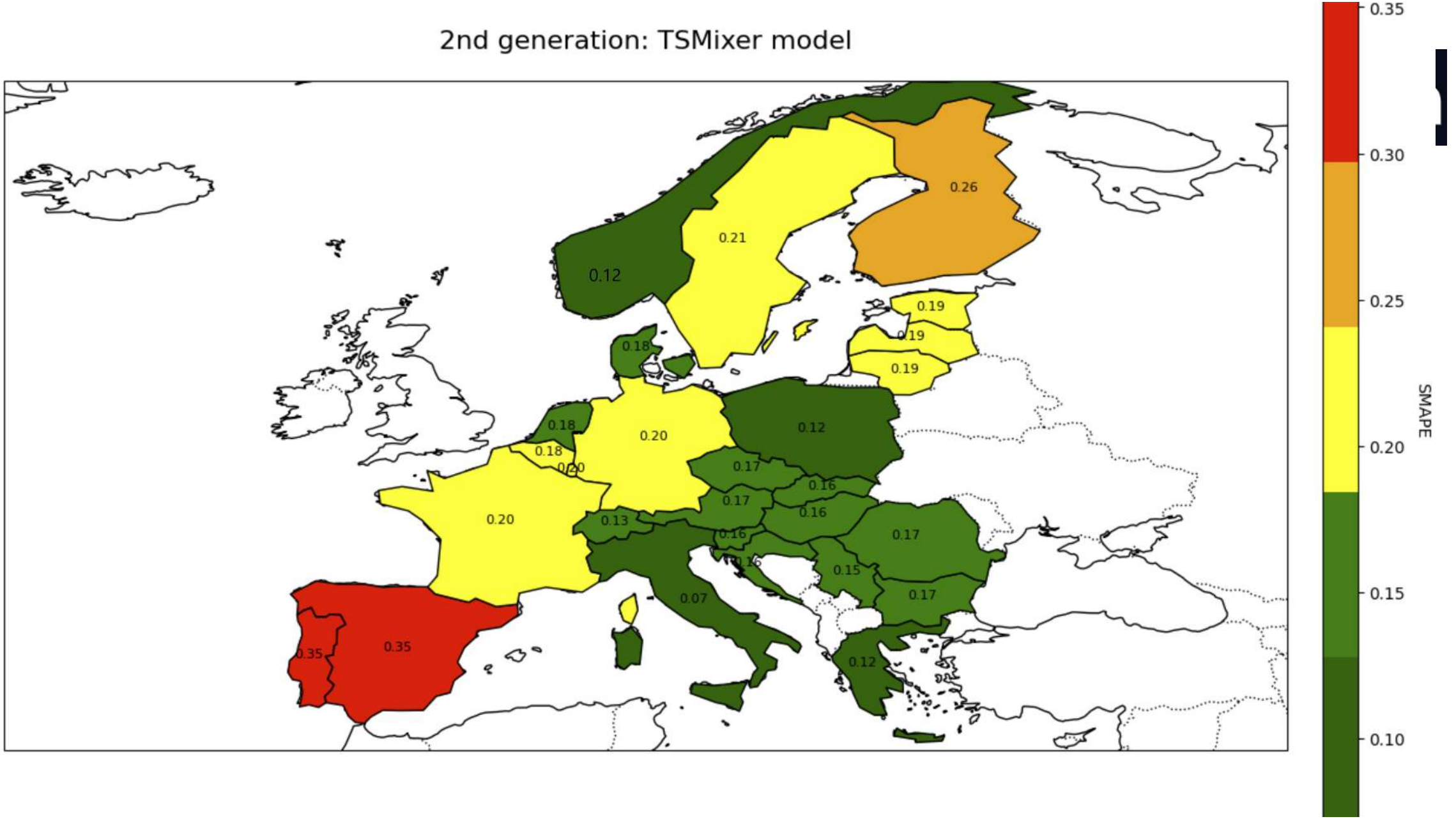


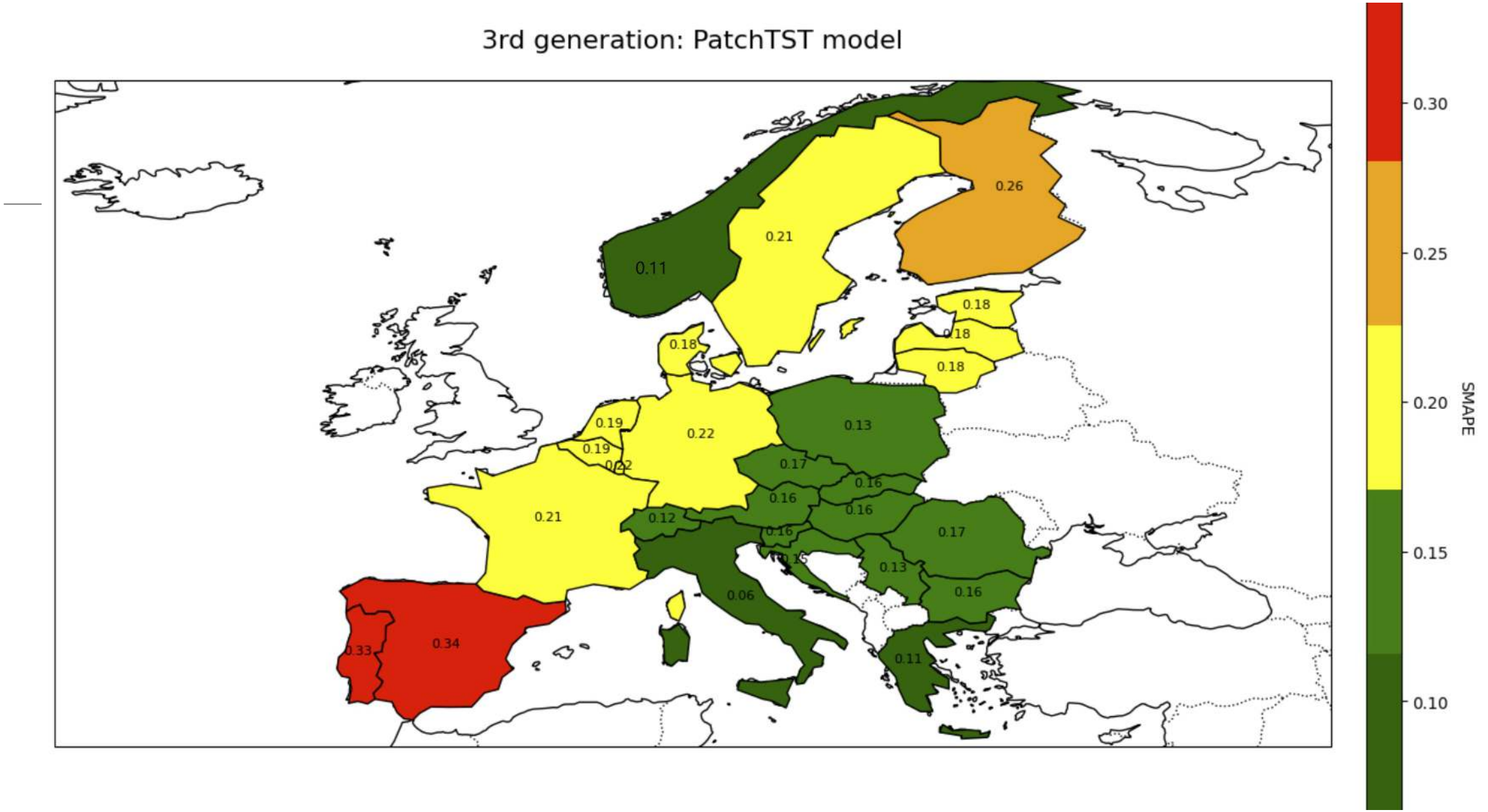
Geomap of average SMAPE achieved by all time series models on the task of electricity price forecasting.

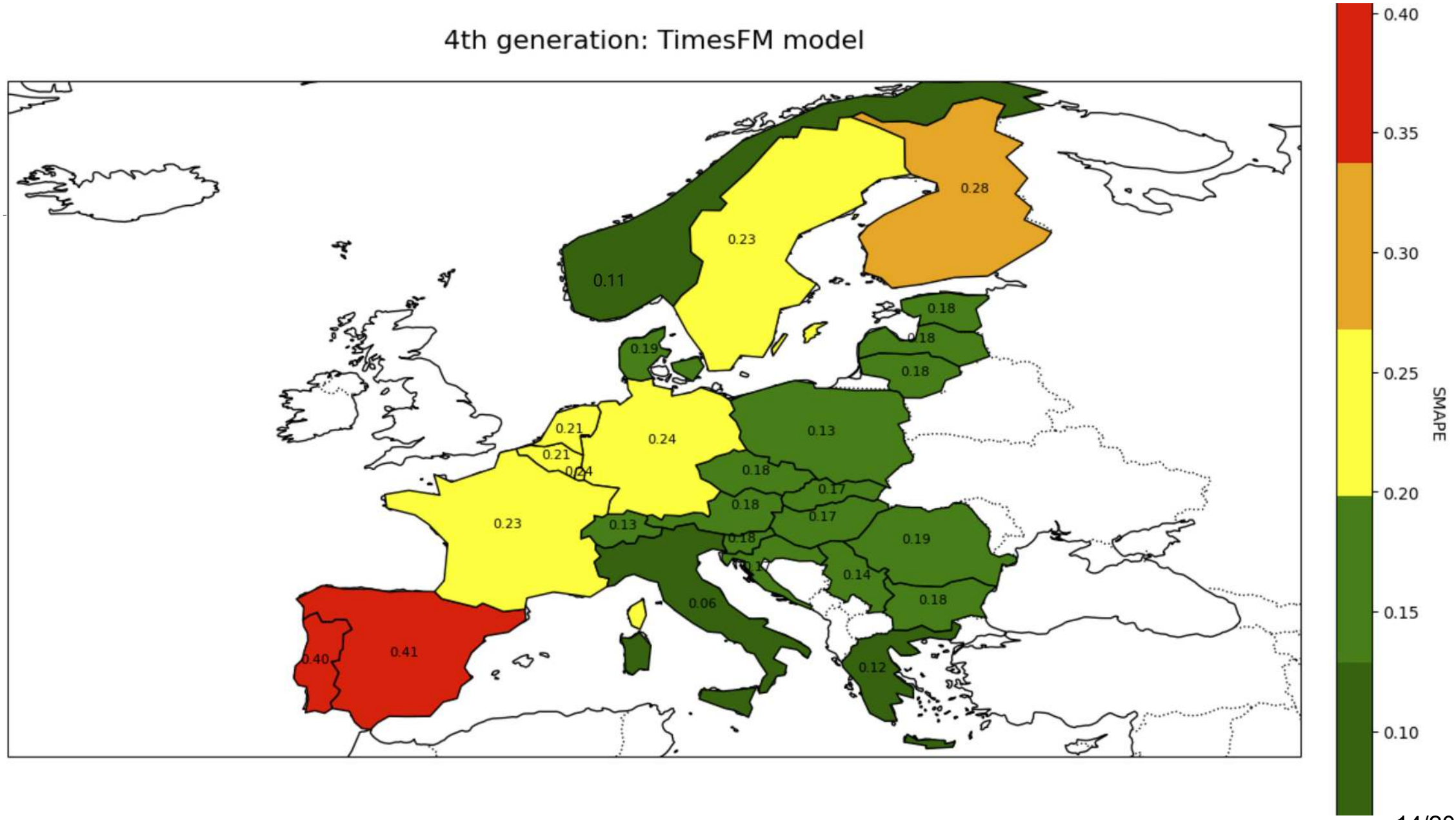
1st generation: ARIMA model

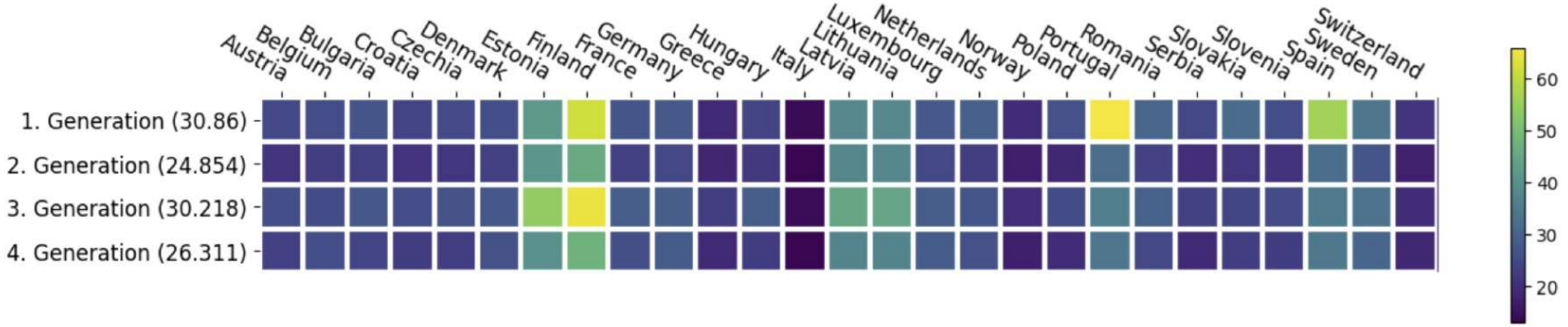


2nd generation: TSMixer model

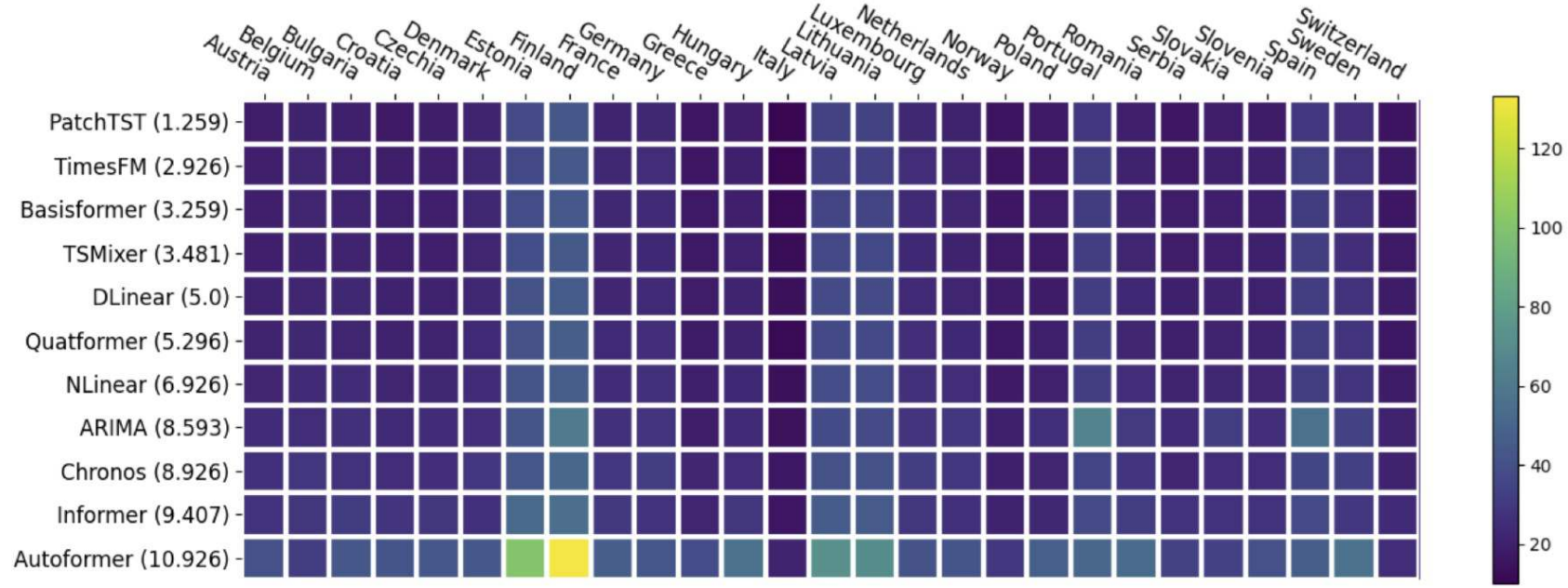



3rd generation: PatchTST model
0.11
0.21
0.26
0.18
0.18
0.18
0.18
0.19
0.19
0.22
0.22
0.13
0.17
0.16
0.16
0.16
0.21
0.12
0.16
0.16
0.17
0.13
0.06
0.16
0.33
0.34
0.11
SMAPE
0.30
0.25
0.20
0.15
0.10


4th generation: TimesFM model
0.11
0.23
0.28
0.18
0.18
0.18
0.19
0.21
0.24
0.13
0.21
0.24
0.18
0.17
0.18
0.17
0.23
0.13
0.18
0.19
0.14
0.18
0.06
0.40
0.41
0.12
SMAPE
0.40
0.35
0.30
0.25
0.20
0.15
0.10



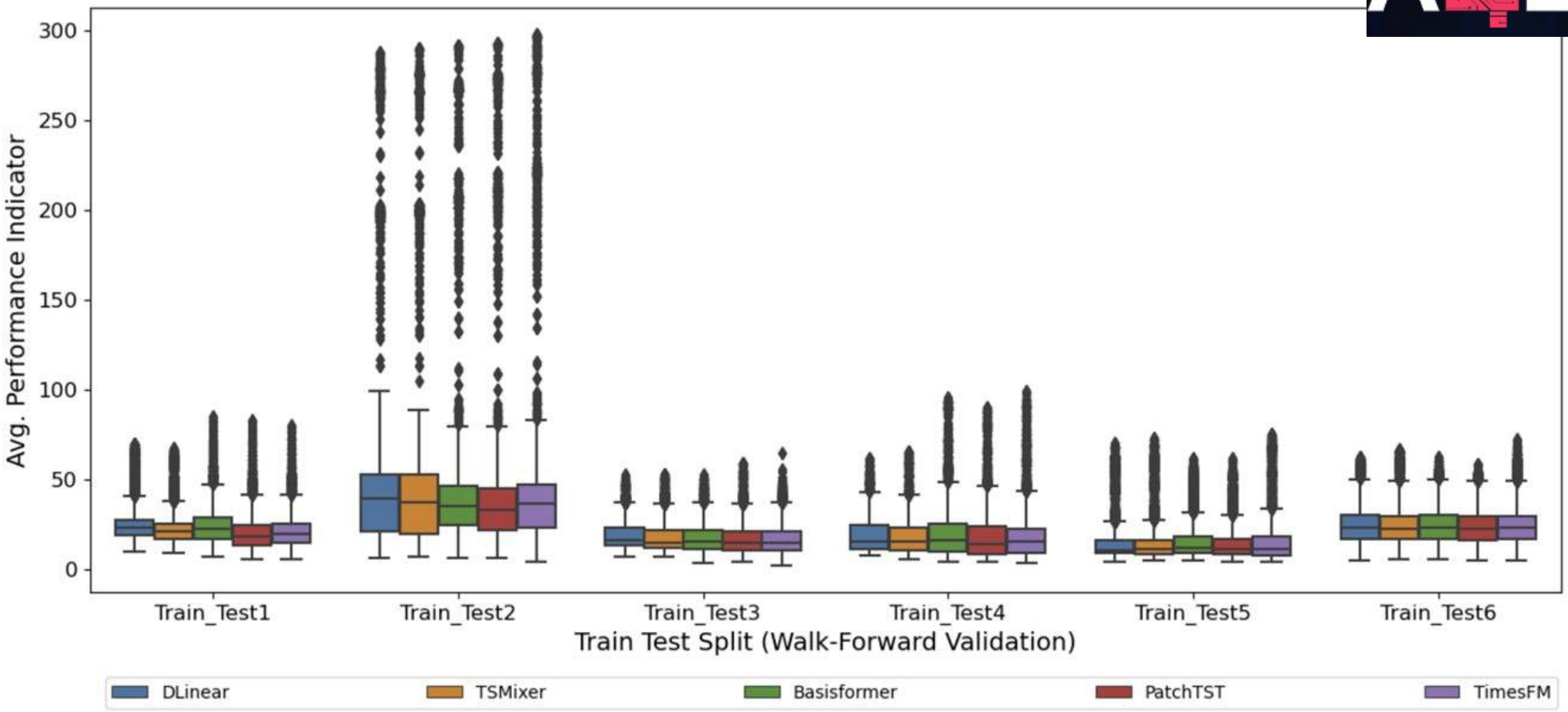
(a) Heatmap of average performance indicator values achieved by model generation.



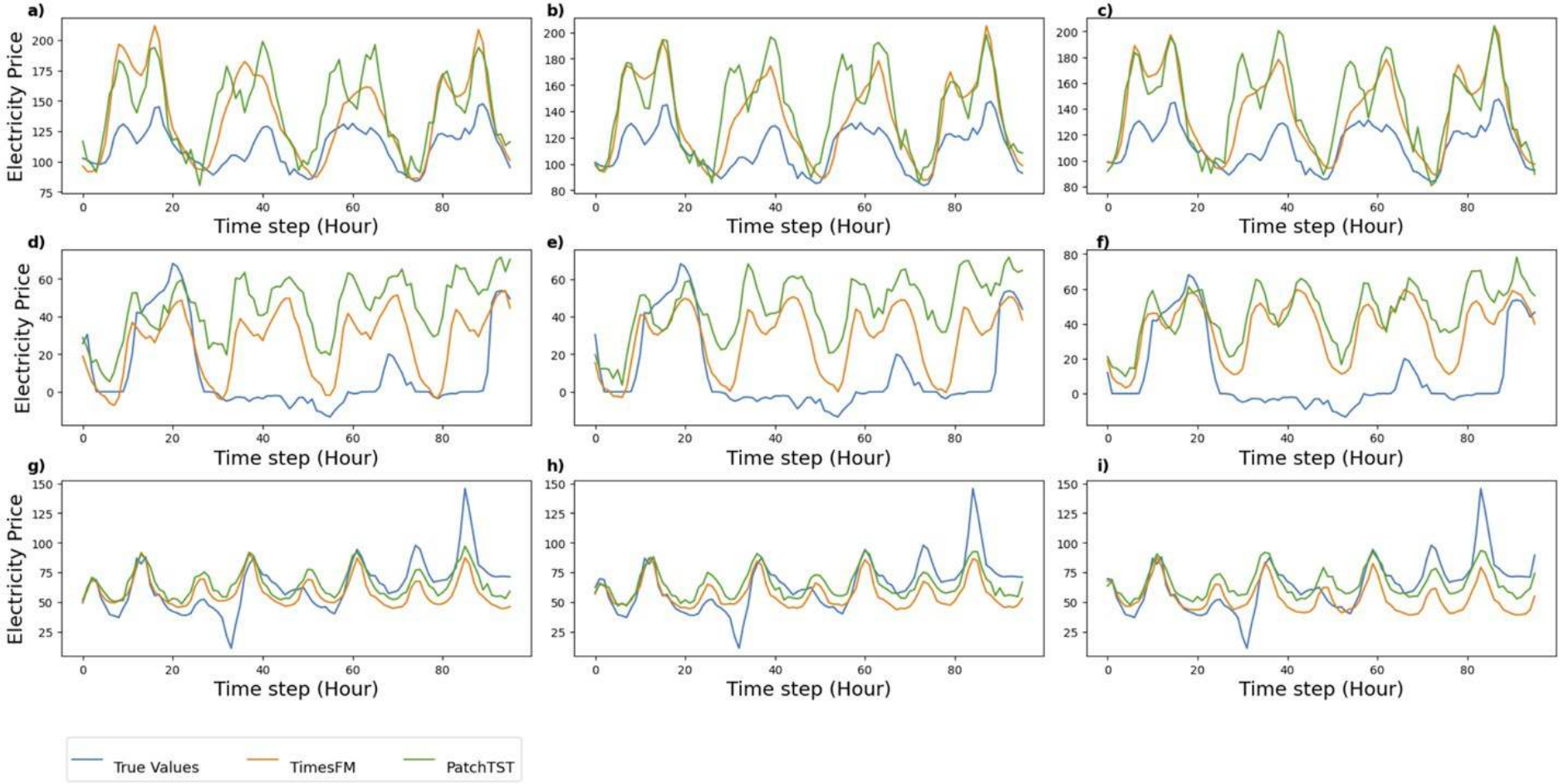
(b) Heatmap of average performance indicator values achieved by each time series model per EU country.

	ARIMA	Autoformer	Basisformer	Chronos	DLinear	Informer	NLinear	PatchTST	Quatformer	TSMixer	TimesFM
Austria	8	11	3	9	5	10	7	1	6	2	4
Belgium	8	11	4	10	3	9	7	1	6	2	5
Bulgaria	8	11	3	9	6	10	7	1	5	4	2
Croatia	8	11	3	9	6	10	7	1	5	4	2
Czechia	8	11	3	9	5	10	7	1	6	4	2
Denmark	8	11	5	10	4	9	7	1	6	2	3
Estonia	7	11	3	9	5	10	8	2	6	4	1
Finland	10	11	3	8	5	9	6	1	7	4	2
France	8	11	4	9	3	10	7	1	6	2	5
Germany	9	11	4	10	3	8	7	1	6	2	5
Greece	7	11	4	10	6	9	8	1	5	3	2
Hungary	8	11	3	9	6	10	7	1	5	4	2
Italy	8	11	3	10	6	9	7	2	4	5	1
Latvia	7	11	3	9	6	10	8	2	5	4	1
Lithuania	7	11	3	9	6	10	8	2	5	4	1
Luxembourg	8	11	4	10	3	9	7	1	6	2	5
Netherlands	9	11	4	10	3	8	7	2	6	1	5
Norway	8	11	3	9	7	10	6	2	4	5	1
Poland	10	11	5	8	4	9	7	2	6	3	1
Portugal	11	10	2	8	4	9	5	1	3	6	7
Romania	9	11	3	8	6	10	7	1	5	4	2
Serbia	9	11	3	8	6	10	7	1	5	4	2
Slovakia	10	11	3	8	5	9	7	1	6	4	2
Slovenia	9	11	3	8	5	10	7	1	6	4	2
Spain	11	10	2	8	5	9	6	1	3	4	7
Sweden	10	11	3	9	5	8	7	1	6	2	4
Switzerland	9	11	2	8	7	10	6	1	4	5	3

Ranking Table



Robustness analysis of top five best performing time series models using walk-forward validation



True vs. Predictions obtained from PatchTST and TimesFM on Germany, a), b), c): test subset 1 with the first three sequence samples, d), e), f): test subset 2 with the first three sequence samples, g), h), i): test subset 6 with the first three sequence samples,

Outlook/Conclusion

Transformers (PatchTST) and foundation models (TimesFM) lead; gains are model-specific, not uniform across the 4th generation

PatchTST performs significantly better than other models reviewed

Future research:

- Expand the application to other domains of energy forecasting
- Investigating the transparency and interpretability of these models
- Provide a data pipeline to realtime forecasting values

Thank you!

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✔https://github.com/QuantLet/AI4EFin.git